Exp 4 MATLAB CODE

```
clc;
clear;
close all;

Fs = 200000;
t = 0:1/Fs:0.02;

fm = 1000;
fc = 20000;
IF = 5000;
flo = fc + IF;

m = cos(2*pi*fm*t);

mu = 0.7;
s = (1 + mu*m).*cos(2*pi*fc*t);

[b1,a1] = butter(4,[18000 22000]/(Fs/2),'bandpass');
rf = filtfilt(b1,a1,s);

lo = cos(2*pi*flo*t);
mixed = rf.*lo;

[b2,a2] = butter(4,[4000 6000]/(Fs/2),'bandpass');
if_signal = filtfilt(b2,a2,mixed);

env = abs(hilbert(if_signal));
```

```
[b3,a3] = butter(4,2000/(Fs/2),'low');
recovered = filtfilt(b3,a3,env);

recovered = recovered - mean(recovered);
recovered = recovered/max(abs(recovered));

figure;

subplot(5,1,1);
plot(t,m);
title('Original Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,2);
plot(t,s);
title('AM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,3);
plot(t,mixed);
title('Mixer Output');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;


subplot(5,1,4);
plot(t,if_signal);
title('Intermediate Frequency Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(5,1,5);
plot(t,recovered);
title('Recovered Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

